

Department of Health and Human Services (HHS) Request for Information (RFI) on the Use of AI in Clinical Care.

1. What are the biggest barriers to private sector innovation in AI for health care and its adoption and use in clinical care?

The American Nurses Association (ANA) believes the biggest barriers to private-sector innovation and adoption of AI in clinical care include fragmented and low-trust data environments that hinder interoperability, unclear and non-harmonized regulatory expectations, and misaligned payment incentives that fail to reward proven improvements in quality or burden reduction. Additionally, concerns about safety, bias, and accountability combined with limited workforce readiness and inadequate nurse engagement in AI governance create cultural and operational resistance. Finally, usability gaps and poor workflow integration often lead to alert fatigue and erode clinician trust, slowing widespread adoption.

Some specific areas where there are barriers to innovation include:

1. **Safety, Privacy, and Ethics:** ANA has foundational concerns with AI innovation in clinical care, including the lack of shared standards for safety, privacy, ethics, and transparency. This leads to uncertainty about reliability, model drift, and potential for biased or incomplete datasets to perpetuate inequities.
2. **Clinical Integration and Validation:** Some barriers to AI innovation in clinical care include lack of clinical input during design, misalignment with clinical workflow, insufficient clinical validation, and limited ability for health systems to test, support, and manage risks of AI technology.
3. **Transparency and Trust:** Vendors often treat algorithms as proprietary, preventing clinicians from understanding output generation, undermining trust, and slowing adoption.
4. **Structural and Economic Barriers:** Significant hurdles include development cost, limited IT/Informatics staffing in smaller hospitals, reliance on electronic health record (EHR) vendors for AI, data quality/interoperability issues, and a lack of clear regulatory expectations and governance frameworks.
5. **Adoption Risks:** Providers using personal AI technology outside of approved software could create patient privacy and cybersecurity risks.

2. What regulatory, payment policy, or programmatic design changes should HHS prioritize to incentivize the effective use of AI in clinical care and why? What HHS regulations, policies, or programs could be revisited to augment your ability to develop or use AI in clinical care? Please provide specific changes and applicable Code of Federal Regulations citations.

ANA has identified the following regulatory and policy changes that HHS can prioritize to further incentivize effective use of AI in clinical care:

1. **Safety and Efficacy Incentives:** HHS should prioritize changes that reward safe, transparent, and workflow-aligned use of AI. This change should be contingent on a mature digital foundation, data quality, and governance structures capable of mitigating AI risks, not just mere adoption. ANA also suggests a testing process proving rigor prior to implementation.
 2. **Regulatory Updates:** Regulations like the Office of the National Coordinator (ONC) Health IT Certification Program (45 CFR Part 170) should be updated to include criteria for AI-enabled decision support. This update should cover transparency, bias testing, explainability, and post-deployment monitoring. ANA also suggests that the Health Insurance Portability and Accountability Act (HIPAA) Privacy and Security Rule (45 CFR Part 164) could be refined to address secondary data use for AI training and vendor data retention.
 3. **Payment Policy Alignment:** CMS should align the Quality Payment Program (42 CFR Part 414) with AI that demonstrably improves outcomes, reduces documentation burden, or enhances safety, explicitly recognizing reductions in cognitive load and alert fatigue.
 4. **Programmatic Support and Oversight:** ANA encourages HHS to support standardized AI governance frameworks and clinical AI evaluation toolkits, emphasizing nursing and frontline clinician involvement. Furthermore, Conditions of Participation (42 CFR Part 482) should clarify expectations for human oversight, incident reporting, and managing AI model drift and health safety events.
 5. **Compliance and Unapproved Tools:** Including the use of any AI product not officially approved as Certified EHR Technology into regulatory and HIPAA requirements could help enforce compliance.
3. **For non-medical devices, we understand that use of AI in clinical care may raise novel legal and implementation issues that challenge existing governance and accountability structures (e.g., relating to liability, indemnification, privacy, and security). What novel legal and implementation issues exist and what role, if any, should HHS play to help address them?**

ANA recognizes that there can be legal and implementation issues regarding AI usage in clinical care, which include the following: unclear liability and indemnification when AI augments clinical decisions; privacy concerns around secondary data use for model training; and risks of bias and discrimination that challenge civil rights compliance. These gaps create uncertainty for providers and vendors. HHS should play a proactive role by issuing guidance on accountability frameworks, privacy-by-design standards, and bias mitigation. Additionally, HHS can promote standardized incident reporting and require transparency and post-deployment monitoring in procurement contracts to ensure safe and equitable AI adoption.

Specific areas of concern include:

1. **Reliable patient notification of when AI is “listening:”** Organizations cannot depend on individual practitioners to consistently and correctly disclose AI use to patients and families in real time. Verbal notification varies by person, workflow pace, and setting, and it breaks down most in high-acuity situations, which is the very environment where ambient documentation or decision-support tools are the most useful. Patients also face a practical transparency problem—they cannot easily tell when an AI tool is actively capturing, transcribing, or analyzing their information versus when it is “off.” Blanket organizational

statements (“we use AI”) do not solve this issue as they do not provide meaningful, situational awareness and can fail to meet expectations for informed decision-making. Guardrails that standardize how and when AI use is communicated—especially for tools that listen in the background—would promote consistency, reduce confusion, and strengthen trust across care settings.

2. **Consent and patient rights in AI-enabled care:** A major gap is the lack of clear, consistent expectations for whether, when, and how patients should be informed about and consent to AI use in their care—particularly when AI influences triage, recommendations, or documentation. Existing privacy frameworks were not designed for algorithmic decision support, automated triage, or predictive analytics, leaving uncertainty about what constitutes adequate notice and meaningful consent. Patients often have no right to know when AI is used, how their data contributes to outputs, or whether their information is used to refine models. This ambiguity undermines trust and creates uncertainty for health systems. A clearer approach would define baseline patient rights: clear notice when AI is involved, plain language explanations of what the AI is doing, and a consent pathway for sensitive uses.
3. **Secondary use of personal health information (PHI) for AI training, development, and vendor retention:** Secondary use of PHI for AI model development, training, refinement, and vendor retention raises distinct governance concerns that current rules address inconsistently. Even where de-identified data is used, modern re-identification risks challenge the adequacy of older de-identification assumptions. Patients and clinicians may not know if data is being stored, repurposed, or used to improve models—particularly when vendors retain datasets or use them across clients. This creates both ethical and operational risk: trust erodes when the data lifecycle is opaque, and organizations face compliance uncertainty when vendor data practices are unclear. Stronger guardrails would emphasize data minimization, defined retention limits, explicit disclosure when data contributes to model training/refinement, and patient-facing clarity about what happens to data after it is collected.
4. **Accountability and liability for AI-influenced clinical decisions:** When AI plays a role in clinical care, accountability can become blurred, especially if the tool is not regulated as a medical device. As malpractice claims emerge, it may be difficult to isolate the AI’s contribution to decision-making, identify whether the clinician reasonably relied on the output, and determine what obligations the organization and vendor had to ensure performance and transparency. Clinicians can be held responsible for decisions influenced by systems they cannot fully evaluate, while vendors may attempt to disclaim responsibility via contract language. This mismatch creates legal and ethical tension and may chill adoption or push unsafe, inconsistent workarounds. A shared-accountability model would clarify roles: what clinicians must verify, what organizations must govern and monitor, and what vendors must disclose and support when their tools influence care.
5. **Transparency requirements for AI tools and vendors:** Transparency is a prerequisite for trust and safe use. Without clear expectations, patients may not understand when AI is part of their care, and clinicians may not know the limits of a tool, what data sources drive outputs, or what conditions degrade performance. Transparency also matters for auditing; if outcomes are questioned, organizations need traceability (what tool version was used, what data inputs were considered, and how outputs were presented). Vendor transparency expectations—such as clear documentation, known limitations, update notices, and auditability—help reduce variability across health systems and prevent “black box” adoption that shifts risk onto frontline clinicians and patients.

6. **Continuous monitoring, drift, bias, unintended consequences:** Traditional governance structures often treat clinical tools as relatively static, but AI systems can change in performance over time as populations shift, workflows evolve, or models are updated. Without a requirement or norm for ongoing monitoring, tools may drift, embed bias, or introduce unintended consequences without being detected promptly. Continuous monitoring is therefore a core patient safety issue, and not just an IT issue. Practical guardrails would normalize lifecycle oversight: routine performance checks, bias surveillance, post-deployment validation, incident reporting pathways, and clear triggers for retraining, rollback, or removal—shared across vendors and health systems.
 7. **AI-specific privacy and security risks:** Current security expectations do not consistently address risks that are uniquely amplified by AI integration, including algorithmic vulnerabilities, training-data poisoning, and risks introduced by large-scale model connectivity and automation. As AI becomes embedded in clinical workflows, security failures can compromise not only confidentiality but also clinical integrity, through manipulated inputs, corrupted training signals, or altered outputs. Modernized guidance would help organizations implement safeguards proportionate to AI risk: tighter access controls, robust logging and audit trails, model/data integrity checks, and vendor requirements that address AI-specific threat models. Standardizing these expectations would reduce variability across organizations and strengthen baseline protections for patients.
 8. **Explicit consent for feeding patient data into AI models:** Consent becomes especially important when patient data is used beyond direct care, such as training, fine-tuning, or improving an AI model. Patients must have clear, understandable information about whether their data will be used to develop or refine algorithms, how long it will be retained, who will have access, and whether it can be used across clients or products. Even when de-identification is claimed, patients may reasonably expect transparency and choice given re-identification risks and the sensitivity of health data. A practical guardrail approach would separate “AI used to support my care today” from “my data used to improve AI for future use,” with clear consent/opt-out mechanisms for the latter whenever feasible.
4. **For non-medical devices, what are the most promising AI evaluation methods (pre- and post-deployment), metrics, robustness testing, and other workflow and human-centered evaluation methods for clinical care? Should HHS further support these processes? If so, which mechanisms would be most impactful (e.g., contracts, grants, cooperative agreements, and/or prize competitions)?**

ANA recognizes that there are promising evaluation methods, metrics, and other considerations for AI use in clinical care, including:

1. **Evaluation Approaches:** The most promising evaluation approaches treat AI as a clinical intervention, not just a predictive model, and they should include frameworks assessing capability, utility, and adoption. Post-deployment evaluations should be prospective and measure changes in clinician behavior and patient outcomes, utilizing trials, A/B testing, stepped-wedge designs, and continuous monitoring for issues like model drift and bias.
2. **Pre-deployment Technical Validation:** These should include metrics similar to clinical trials, calibration, fairness metrics, and robustness testing (e.g., performance under data shifts, missing data, and adversarial scenarios). External validation on data from multiple sites and populations is essential.

3. **Workflow and Human-Centered Evaluation:** Methods like usability testing, simulation-based assessments, time-motion studies, and cognitive load measures are important for identifying how AI fits into clinical workflows, reduces burden, and supports clinical judgment. Evaluating explainable AI is also key to ensuring clinicians can understand, trust, and act on outputs.
4. **Specific Metrics:** ANA's suggested metrics include time saved documenting and increased time at the bedside, readmission rates to evaluate discharge planning and education, surgical site infection rates, and timely clinical data in the chart.
5. **HHS Support Mechanisms:** The most impactful mechanisms for HHS to support these processes are targeted grants and cooperative agreements, contract and certification guidance, and support for learning collaboratives and shared evaluation infrastructure. Education on AI use and documentation should be required at the school/education level rather than per healthcare organization.

5. How can HHS best support private sector activities (e.g., accreditation, certification, industry-driven testing, and credentialing) to promote innovative and effective AI use in clinical care?

As HHS seeks to strengthen private-sector efforts, ANA urges the agency to endorse standardized accreditation and certification frameworks for healthcare AI that incorporate transparency, bias mitigation, and safety criteria. Additionally, HHS should fund nurse-focused AI education and micro-credentialing programs through ANA and the American Nurses Credentialing Center (ANCC) to build workforce readiness. These steps will promote confidence, ensure ethical deployment, and position nurses as key partners in advancing effective AI adoption across clinical care.

1. **Support for Private Sector Efforts:** HHS should recognize and endorse trustworthy private frameworks (accreditation, certification, industry-driven testing) without mandating a single model, and use funding/procurement levers to prioritize vendors and health systems using accredited/certified AI tools.
2. **Certification and Vendor Requirements:** Certification of AI tools is desired to ensure a certain level of rigor in testing, and HHS should work with EHR vendors should be required to meet Certified Technology Requirements to assist smaller organizations with limited resources.
3. **Infrastructure and Training:** HHS must support shared infrastructure and learning collaboratives to lower the barrier for smaller organizations, and require clinicians to be trained within their licensure for safe and accurate technology use.
4. **Promote AI Use and Human-Centered Design:** HHS can promote the use of AI in clinical care, such as "smart" hospital rooms, and explicitly center human-centered design and workflow fit in any supported accreditation or certification efforts.

6. Where have AI tools deployed in clinical care met or exceeded performance and cost expectations and where have they fallen short? What kinds of novel AI tools would have the greatest potential to improve health care outcomes, give new insights on quality, and help reduce costs?

ANA acknowledges that some AI tools have exceeded expectations in areas like ambient documentation and workflow assistants, which, according to those companies, significantly reduce charting time and improve usability. This includes imaging, where validated models

enhance diagnostic accuracy. However, many tools have fallen short due to poor workflow integration, alert fatigue, and lack of transparency, leading to low clinician trust. Importantly, we lack strong evidence that these tools reduce workload or mitigate burnout, underscoring the urgent need for standardized outcome measures that capture clinician burden and well-being. Looking forward, the greatest potential lies in nurse-centric solutions such as explainable triage and risk stratification, ambient nursing documentation with reasoning support, staffing optimization tools that improve well-being, and AI-driven quality measurement systems designed to reduce burden while improving patient outcomes.

Specific examples include:

1. **Success of AI Tools:** Examples include AI-assisted imaging for faster turnaround times, capacity and throughput tools for bed management, and operational optimizations. Mobile Point of Care Documentation devices for nursing have also greatly improved bedside care, as well as ambient listening for provider documentation.
2. **Shortcomings of AI Tools:** AI tools fall short when they are generic, layered into already fragile systems, increase alert fatigue, add clicks, provide unclear recommendations without explainability, or do not translate into better outcomes due to a lack of underlying workflows/resources. Some predictive models, models trained on biased data, or tools that reinforce existing inequities have also fallen short. A lack of clinician input in development, particularly for nursing workflows, can also lead to tools that don't improve workflows and result in passed-on costs to patients without clear benefits.
3. **Greatest Potential for Novel AI Tools:** The greatest potential lies in tools that directly support patient safety, nursing workflow, and proactive care, such as patient-movement and safety monitoring tools (e.g., detecting unsafe movement, predicting fall risk). Other high-potential tools include ambient and workflow-aware documentation support for nurses, AI-enabled care coordination and handoff tools, and operational/staffing optimization tools that align staffing with acuity and patient needs in real time. These tools should be deeply embedded in workflows and evaluated as clinical interventions.
4. **Need for Data:** There is a current lack of sufficient data on the performance and cost-effectiveness of AI tools.

7. Which role(s), decision maker(s), or governing bodies within health care organizations have the most influence on the adoption of AI for clinical care? What are the primary administrative hurdles to the adoption of AI in clinical care?

ANA believes that the adoption of AI in clinical care is most influenced by Chief Nursing Officers, nursing informatics leaders, and quality or operations executives, especially when nurses have a formal role in governance. However, there are administrative hurdles such as, unclear ownership of AI initiatives, lengthy procurement processes, and limited time or resources for workforce training, which slow implementation and reduce frontline engagement.

1. **Influential Roles:** Executive sponsorship is mentioned as having an impact on AI adoption.
2. **Cost and Resources:** The most significant barriers include the cost of AI add-on solutions and the lack of implementation resources, with vendors shifting costs and risk to organizations.
3. **Governance and Accountability:** Primary administrative hurdles stem from fragmented governance, unclear accountability, and a lack of foundational digital maturity, leading to

prolonged contracting and legal review due to the absence of a consistent framework for addressing data use, liability, and vendor accountability.

4. **Government Role:** HHS can best support adoption by establishing clear, consistent expectations for safety, transparency, and accountability, and by defining minimum criteria for responsible clinical AI, such as bias testing and monitoring, while vendors build accreditation and testing programs.

8. Where would enhanced interoperability widen market opportunities, fuel research, and accelerate the development of AI for clinical care? Please consider specific data types, data standards, and benchmarking tools.

ANA urges HHS to provide incentives to accelerate and support the development of AI for clinical care in the following ways:

1. **Data Interoperability:** Enhanced interoperability would significantly widen market opportunities and accelerate AI development by improving access to high-quality, standardized clinical data, which is essential for reliable AI model training and safer deployment.
2. **High-Value Data Types:** Prioritizing interoperability for data types such as vital signs, laboratory results, medication administration records, device data, care plan, nursing assessment, imaging metadata, and patient-generated health data is crucial.
3. **Standards and APIs:** Greater adoption of fast healthcare interoperability resources (FHIR)-based application programming interfaces (APIs), United States Core Data for Interoperability (USCDI) expansion, and consistent implementation of health level (HL7) standards would reduce variability and allow vendors to build tools that work across various systems.
4. **Benchmark Datasets:** Shared, de-identified benchmark datasets built on common data models like observational medical outcomes partnership (OMOP) would accelerate research by allowing models to be evaluated against consistent reference standards.
5. **Specific Areas:** Enhanced interoperability for device and IT data is important for AI tools focused on patient safety, mobility monitoring, and early deterioration detection, and secure databases of de-identified health information could also help.

9. What challenges within health care do patients and caregivers wish to see addressed by the adoption and use of AI in clinical care? Equally, what concerns do patients and caregivers have related to the adoption and use of AI in clinical care?

ANA knows that patients and caregivers want AI to address challenges like reducing delays in care, improving communication, and freeing clinicians from excessive documentation so they can spend more time delivering care. At the same time, they have concerns about privacy, data misuse, bias, and the risk that over-automation could erode human interaction and accountability. Transparent communication and strong nursing involvement are essential to building trust and ensuring AI adoption aligns with patient-centered values.

Specific challenges include:

1. **Desired AI-Addressed Challenges:** Patients and caregivers want AI to address safety risks, care coordination gaps, long wait times, navigating the complex system, and preventing

avoidable harm. They also desire improvements in communication between clinicians, reduced delays in diagnosis, ensuring nothing is missed in care, better access to primary care, and addressing clinician burnout.

2. **Patient Education/Understanding:** A major challenge is the general public's lack of education and skepticism regarding AI, particularly in rural areas with poor health and technology literacy, which hinders understanding of how data is used and how AI functions.
3. **Concerns about AI:** Concerns center around privacy, consent, the potential for AI to replace human judgment, reduce relational aspects of care, introduce bias or inequity, and create new errors that are difficult to detect. Patients also worry about the accountability of clinicians for algorithm-influenced decisions.
4. **Organizational/Systemic Challenges:** Small hospitals with limited knowledge and resources struggle to prioritize and afford updated AI technology, and organizations cannot solely take on the responsibility of re-educating the public.
5. **Social Determinants of Health (SDOH) and Resources:** AI is desired to help link patients with resources related to SDOH challenges to ensure they can attend appointments and follow up on prescribed care.

10. Are there specific areas of AI research that HHS should prioritize to accelerate the adoption of AI as part of clinical care?

ANA urges HHS to prioritize research on standardized outcome measures to assess AI's impact on clinician workload, burnout, and equity, as current evidence is limited. Additional focus areas include safety and bias mitigation, cost-benefit analyses, and real-world effectiveness studies through nurse-led trials.

1. **Prioritized AI Research:** HHS should focus on AI research that evaluates real-world impact, safety, equity, and burden reduction, including prospective studies of AI as a clinical intervention, human factors, workflow impact, economic evaluation of implementation/maintenance, and systematic bias assessment.
2. **Need for Further Literature/Studies:** The existing literature is considered lacking, necessitating more long-term studies to understand the impact and return on investment, as well as a need for more rigorous, standardized economic evaluations that capture indirect costs like workflow redesign, training, governance, and monitoring.
3. **Existing Literature Findings:** A growing body of literature indicates that AI can improve diagnostic accuracy, workflow efficiency, and patient outcomes, but many studies are limited, specialty-specific, often optimistic, and inconsistent in reporting safety events, equity impacts, or long-term effectiveness.
4. **Implementation Considerations:** It is important to promote mobile point-of-care implementation and avoid a "one-size-fits-all" approach, as each discipline and setting will have unique needs, requiring consideration of change management and the mental load on clinicians.